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PPOL 6801: Text as Data
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Final Project Proposal

The Trump administration recently put out an Executive Order demanding that all Diversity, Equity and Inclusion (DEI) initiatives and programs must be eliminated throughout the federal government (*Ending radical and wasteful government dei programs and preferencing*, 2025). This order has led to the executive branch flagging specific programs for potential violations by looking at vocabulary used; however, some of these "violations" for involving DEI have been wrongly accused of being so and caught attention from the public. For example, the Enola Gay airplane which was the plane that dropped the first atomic bomb in Japan in World War II had records flagged for removal (Copp et al., 2025). This flagging does not seem to align with the mission of this Executive Order and hence sparks inspiration generally for my final project. I want to look into congressional records and bills to understand the context of the words, "diversity", "equity", and "inclusion" are specifically used. Furthermore, I want to analyze the differences in use across parties and understand the topics created surrounding these words and the documents they appear in.

Data Sources

I plan to use data from Congress.gov to obtain both congressional records and bills. These records will be filtered to only include those that include the words of interest, "diversity", "equity", and "inclusion." Additionally, I will have to determine a timeline that is appropriate to give enough records, but also to limit the number of results to a reasonable number if there are too many. The metadata from the data download will also be very helpful in structural topic modeling to utilize author/speaker name and the party of those participating in congressional records or writing bills. Furthermore, some bills might have specific labels that could be interesting to conduct a supervised learning task within structural topic modeling.

Data Retrieval Approach

Using the Congress.gov API should be the easiest way to pull all search results with necessary filters, especially if the number of results is quite large, which I imagine will be the case. I will have to set up a developer account to do so and potentially use Python first to pull all necessary data before then loading data into R Studio.

Methods

Preprocessing - First, we will need to properly preprocess the retrieved data to ensure that our analysis is meaningful. While this may be an iterative process involving some discovery while working with the data, we currently anticipate the following preprocessing steps to be necessary:

1. Removal of stopwords
2. Lowercase

3. Removal of numbers, symbols, and punctuation
4. Lemmetizing/Stemming

Key Words in Context - This method will be the first to give some general descriptive statistics on the textual data we are observing. We will look at specific pre- and post-context to understand what types of words appear around the words of interest that we are searching for. We may also consider using this window of words as a subset to explore in topic modeling rather than looking at entire documents.

Topic Modeling: Structured and Keyword-Assisted - We will use structural topic models to involve the metadata in the creation of topics across documents. These topic models will look more generally to see what types of topics emerge from congressional documents using our words of interest. We will primarily focus on author and ideology for the specific metadata used in the structural part of our topic model. These will hopefully help to reveal if party lines follow any patterns when it comes to using the words of interest.

Another topic model to explore is keyword-assisted topic models. We will use the words of interest as keywords to create topics. While this might not be valuable in topic creation, it could provide insight to which words will appear within each of the keyword associated topics.

Word Embeddings - Finally, we will look at word embeddings to try to better understand the meaning of the words of interest in the context of the documents that we are analyzing. While we understand what these words mean from a general stand-point, I am interested in if this differs in the context of congressional conversations and legislative terms. This method will involve various decisions like window size, number of dimensions, and type of algorithm. These decisions will likely be formed through an iterative process to understand which parameters perform best.

Definition of Success

This project will be considered successful if there are clear insights drawn on how the words of interest are used in context. If diversity, equity, and inclusion are used in the context that most people would imagine them to be used in, then that is interesting, but we also will want to look at specific outliers to understand the more rare ways that these words are used and see if they would be supported or not by the current administration. Additionally, understanding the topics that emerge from the documents that use these words to also see if those align with what is expected or not, and if they align with the current administration's goals or not. Additionally, we will use various validation methods to ensure that the results we achieve are actually viable and reasonable for our analysis. These methods will be both manual, including reading specific documents, and automatic, including coherence or perplexity, in nature.

Sources

Copp, T., Baldor, L. C., & Vineys, K. (2025, March 7). War heroes and military firsts are among 26,000 images flagged for removal in Pentagon's dei purge. AP News.
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